

EVIDENCE-FIRST INDUSTRIAL AI

ProofRAG

Maintenance answers you can verify.

A multimodal troubleshooting assistant that returns cited guidance—or explicitly abstains when the evidence is insufficient.

PROBLEM STATEMENT AND RELEVANCE

Industrial maintenance knowledge is fragmented across long manuals, scanned service bulletins, tables, diagrams, and versioned procedures. During a fault, technicians must determine not only what a document says, but whether it applies to the correct equipment, configuration, and manual revision.

Traditional keyword search misses relationships and exact procedural context. Generic AI assistants may produce plausible instructions without reliable evidence, blend conflicting versions, or cite text that does not support the answer. In a safety-sensitive environment, a fluent answer is not enough: the technician must be able to inspect the source, see its provenance, and recognize when the available documents cannot support a safe conclusion.

ProofRAG addresses this gap with one non-negotiable rule: **no evidence, no answer**. It retrieves applicable document evidence, preserves page and region provenance, presents safety prerequisites before intervention steps, and refuses unsupported questions. It assists qualified personnel while leaving site procedures and human authority in control.

SELECTED THEME

Theme 2
Multimodal Maintenance Intelligence Agent

PRIMARY USERS

Maintenance technicians · Maintenance engineers · Field service · Technical support · Control-room personnel

TEAM

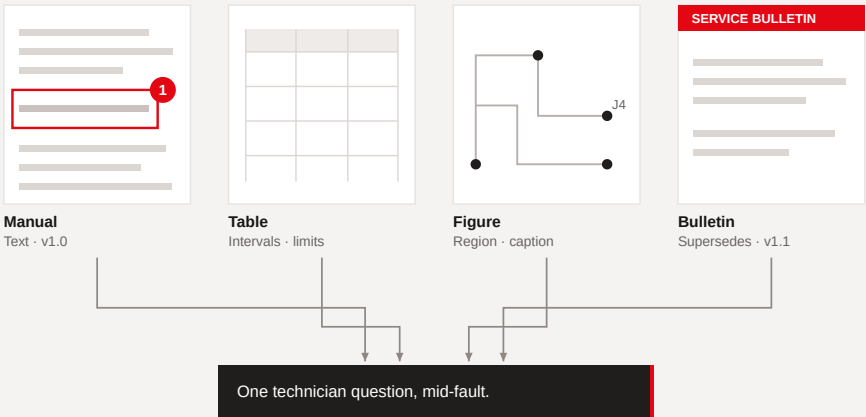
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PRODUCT BOUNDARY

Decision support and source navigation only. No equipment control. No approval of maintenance or return to service.

CATEGORY

Grounded troubleshooting and technical-document intelligence.



The applicable answer is rarely in one place. ProofRAG keeps every fragment traceable to its page and region.

01

Fragmented knowledge

The applicable procedure may be split across a base manual, later bulletin, table, figure, and scanned page.

02

Version risk

A technically correct value can still be unsafe when it comes from an outdated or inapplicable document version.

03

Verification burden

Technicians need the exact page, source type, and highlighted region—not an unsupported summary.

04

Missing evidence

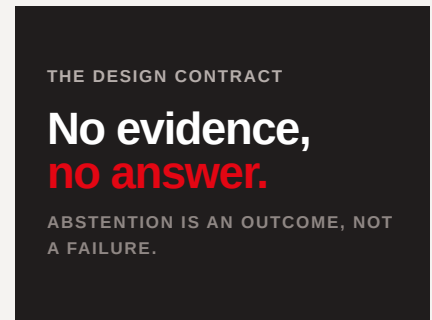
When the corpus cannot answer, the correct product behavior is visible abstention and escalation.

IDEA AND APPROACH

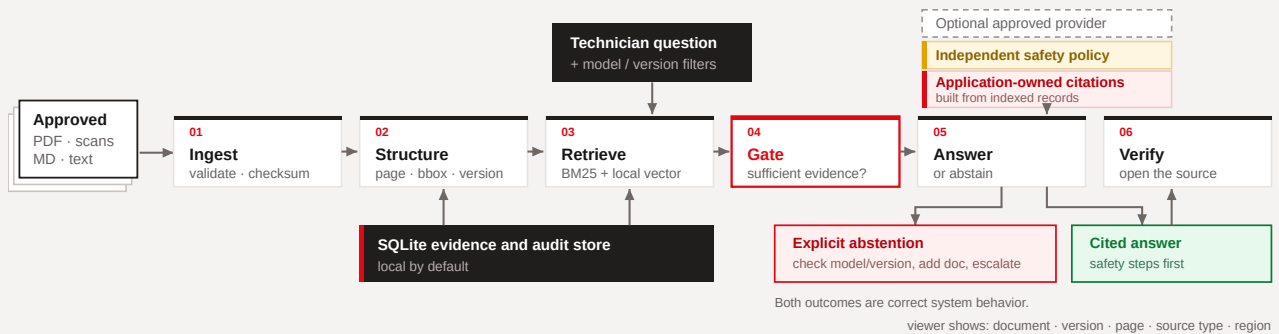
From source material to cited guidance

ProofRAG ingests PDF, scanned, Markdown, and plain-text technical documents and converts them into structured evidence blocks for text, OCR, tables, and figure/caption regions. Each block retains document identity, version, page, source type, and bounding box when available.

At query time, ProofRAG combines exact-term retrieval—important for fault codes, connector names, model identifiers, and numeric thresholds—with a deterministic local vector signal. An evidence-sufficiency gate decides whether an answer is permitted. Supported responses receive application-owned citations and visible safety notices; weak evidence produces an explicit abstention with no procedural citations.



THE CONTROLLED PATH — INGEST, STRUCTURE, RETRIEVE, GATE, ANSWER OR ABSTAIN, VERIFY



WHAT MAKES IT DIFFERENT

Evidence-first answers The answer is downstream of retrieved records and an explicit sufficiency gate.	Application-owned citations Citation IDs, document metadata, pages, source types, and regions come from indexed database records—not generated prose.
Region-level verification PDF bounding boxes support highlighted source previews so users can inspect the exact evidence area.	Version-aware retrieval Manual versions remain visible and filterable; the demo shows a later service bulletin superseding a base interval.
Multimodal evidence model Text, OCR, tables, and figure/caption regions share one retrieval and citation pipeline.	Local-first operation The deterministic extractive path works without sending documents to a hosted AI provider.

ARCHITECTURE LAYERS

A	Ingestion File/type/size validation · SHA-256 duplicate detection · PyMuPDF layout extraction · optional Tesseract OCR hook · table and figure/caption extraction
B	Evidence Bounded chunks · normalized content · hashing embeddings · document/version/page/source metadata · bounding boxes · SQLite persistence
C	Retrieval BM25 exact-term scoring · local vector similarity · weighted score merge · document/model/version filters · weak-evidence pruning
D	Grounding and safety Evidence threshold · extractive or approved OpenAI-compatible provider · application-owned citations · independent safety notices · abstention
E	Experience and audit FastAPI · technician web interface · answer confidence · citation cards · highlighted source viewer · query audit trail

Default extractive mode keeps documents local and requires no hosted model. When an approved provider is enabled it receives only retrieved excerpts, and it never owns citations, safety notices, or the decision to abstain.

IMPLEMENTATION AND IMPACT

Trust is designed into the pipeline

MODEL TYPE

Hybrid lexical retrieval + deterministic local vector baseline + evidence-gated extractive answering.

Optional hosted generation is replaceable and remains downstream of retrieval, grounding, and application-owned citation controls.

TECHNICAL STACK

Backend Python 3.12 · FastAPI · Uvicorn

Documents PyMuPDF · optional Tesseract OCR

Retrieval BM25 · feature hashing · cosine similarity

Storage SQLite evidence and audit store

Interface HTML · CSS · JavaScript

Quality uv · pytest · Ruff · mypy · Docker

DEMONSTRATION PROOF

- 01 Supported fault.** E-17 retrieves lockout/tagout, absence-of-voltage, cooling filter, fan, and connector J4 evidence.
- 02 Version conflict.** A later bulletin changes the dusty-site filter inspection interval from 500 hours to 250 operating hours or monthly.
- 03 Safe abstention.** A refrigerant type and charge question is unsupported and receives no procedural answer.

SYNTHETIC DEMONSTRATION —
NOT ABB EQUIPMENT

INNOVATION HIGHLIGHTS

No-evidence / no-answer contract

Abstention is a first-class product outcome, not a fallback sentence added after generation.

Inspectable spatial provenance

Document page coordinates survive ingestion and retrieval, allowing highlighted source inspection rather than citation-only trust.

Offline by default

The baseline path is deterministic and local; external AI is opt-in and receives only retrieved excerpts.

Citation authority outside the model

The application constructs citations from indexed records, preventing the answer provider from inventing source IDs or provenance.

Safety before intervention

Independent warnings and evidence-derived prerequisites appear before procedural actions.

Conflict and gap visibility

Version filters expose document applicability, while abstentions and query logs reveal missing documentation.

EVALUATION
FRAMEWORKTARGETS — NOT
RESULTS

- ☐ Retrieval Recall@5 and mean reciprocal rank
- ☐ Version accuracy and source-type recall
- ☐ Expected-term coverage
- ☐ Faithfulness, citation precision, citation completeness
- ☐ Unanswerable accuracy and false-answer rate
- ☐ Safety prerequisite ordering
- ☐ Median / P95 latency and indexing throughput
- ☐ OCR warning/failure rate and duplicate behavior

Six smoke cases exist. Expand to at least 50 human-reviewed cases before making competitive performance claims. Runtime verification of the locked environment is still pending.

EXPECTED BUSINESS VALUE

- Shorter navigation time from a fault code to the applicable procedure.
- Fewer unsupported troubleshooting suggestions.
- Better auditability through visible source, version, page, and evidence region.
- Faster transfer of documented knowledge to less-experienced technicians without hiding the source.
- Earlier discovery of missing, conflicting, or outdated documentation.
- A lower-risk adoption path: the first pilot is read-only, bounded to approved documents, with no equipment-control path.

Every item is an expected outcome until measured in a real pilot.

PROTOTYPE TO PILOT

1 Verified prototype

Run the locked environment, fix defects, capture screens, execute tests, and publish measured evaluation output.

2 Bounded pilot

One approved equipment family, immutable document versions, technician-authored questions, human review.

3 Production hardening

Identity and role access, encrypted object storage, PostgreSQL, filtered vector search, sandboxed ingestion, monitoring, retention.

4 Controlled expansion

Dense industrial embeddings, vision-language diagram description, knowledge graphs, applicability rules, feedback workflows.

LIMITATIONS AND FUTURE WORK

- Figure support currently indexes regions and nearby captions; it is not full diagram reasoning.
- OCR depends on source quality and host Tesseract availability.
- The hashing embedder is a reproducible baseline, not a production semantic model.
- Version precedence is user-filtered; automated effective-date and applicability rules are future work.
- Evidence confidence is retrieval-derived and requires calibration on a larger reviewed set.

Production authentication, antivirus/content disarm, and vector infrastructure are out of current scope.

PILOT ASK

Evaluate ProofRAG on one bounded equipment family using approved manuals and technician-authored questions, then expand only after citation and abstention targets are met.

ProofRAG
because the source is
part of the answer.